TOTAL ACYCLICITY IMPLIES RIGHT ACYCLICITY FOR THE SECTION FUNCTORS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the four listed functors.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Totally acyclic sheaf and the criterion by extended Cech complexes. Same chapter, the definition of total acyclicity and the criterion characterising it, which is the companion lemma of this section. Supplies the property of the sheaf assumed in the statement.
- Right acyclicity for a left exact functor. Same chapter together with the general theory of derived functors: a sheaf is right acyclic for a left exact functor when the higher derived functors vanish on it. Supplies the property verified for the four functors listed.

Source reference. Original lemma 03CR: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2224–2224. Navigation pointer invoked at proof line 2224; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 072W: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 747–747. Navigation pointer invoked at proof line 2301; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Lemma 0.1. *Let $(Sh(\mathcal{C}), \mathcal{O}_\mathcal{C})$ be a ringed topos. A totally acyclic sheaf is right acyclic for the following functors:*

1. *the functor $H^0(U, -)$ for any object U of $\mathcal{C}$,*
2. *the functor $\mathcal{F} \mapsto \mathcal{F}(K)$ for any presheaf of sets K,*
3. *the functor $\Gamma(\mathcal{C}, -)$ of global sections,*
4. *the functor f_* for any morphism $f : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ of ringed topoi.*

Proof. Part (2) is the definition of a totally acyclic sheaf. Part (1) is a consequence of (2) as pointed out in the discussion following the definition of totally acyclic sheaves. Part (3) is a special case of (2) where $K = e$ is the final object of $Sh(\mathcal{C})$.

To prove (4) we may assume, by Modules on Sites, Lemma 03CR that f is given by a morphism of sites. In this case we see that $R^i f_*$, $i > 0$ of a totally acyclic sheaf are zero by the description of higher direct images in Lemma 072W. □